Conference proposal

Projet Applicatif 3 · prepared for the common framework requirement (title, abstract, targeted venue)

Proposed title	**Accountable Federated Learning for Cross-Border Anti-Money Laundering: Detecting Poisoners Without Seeing Transactions**
Short title	SENTINEL-FL
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Type	Short paper (8–10 pages) or workshop paper, with artefact (open repository and demonstration)
Keywords	federated learning; anti-money laundering; differential privacy; model poisoning; smart contracts; auditability; GDPR

Abstract

Banks must detect money laundering that spans institutions and borders, but data protection law, banking secrecy and the AML tipping-off prohibition prevent them from pooling transaction data. Federated learning lets banks train a shared model without moving data, yet a consortium also needs assurance that no participant leaked or poisoned anything, and evidence that a supervisor can check. We present SENTINEL-FL, a federated AML consortium that combines three layers: cryptographic participation (Ed25519-signed updates and a smart-contract registry that decides who may contribute), statistical robustness (update anomaly scoring, quarantine and robust aggregation), and accountability (a signed hash-chain ledger whose per-round Merkle roots are anchored on-chain, with a regulator view). Each bank trains with DP-SGD under a capped Rényi-DP budget. We argue that secure aggregation conflicts with poisoner identification in small consortia and adopt visible but privatised updates instead. On synthetic AMLworld-like data with three non-IID banks, the federated model reaches PR-AUC 0.859 ± 0.028, on par with centralised training (0.840). A label-flipping bank drives an undefended model to 0.012; with our defences the model keeps 0.835 and the attacker is quarantined in its second malicious round. Backdoor success drops from 1.0 to 0.0, exfiltration and tampered updates are rejected by protocol, and an incident is reconstructed from the ledger and on-chain anchors, with an insider edit located at the exact record. Privacy costs utility: PR-AUC 0.703 at $\varepsilon \approx 6$. The prototype runs on Flower, PyTorch and Opacus.

Contributions

- A requirements analysis mapping GDPR, AML and DORA obligations to federated-learning controls, including alert minimisation for the tipping-off rule.
- An accountable FL design that identifies and quarantines poisoners, with access control and round anchoring in a smart contract, and an analysis of why secure aggregation hinders this in small consortia.
- An open prototype (bank and aggregator nodes, regulator view, Flower + PyTorch + Opacus engine) and an evaluation of the five test scenarios: poisoning, privacy budget, leakage, audit reconstruction, comparison with centralised training.

Targeted venue

Venue	Why it fits	Fit
DPM — International Workshop on Data Privacy Management, co-located with ESORICS (primary)	Privacy engineering with regulatory grounding; accepts prototype-based work; European audience	Primary

Venue	Why it fits	Fit
ARES — International Conference on Availability, Reliability and Security (and its workshops)	Applied security, trust and auditability; European; short papers	Alternative
C&ESAR — Computer & Electronics Security Applications Rendezvous (France)	French applied-security community, industry and government audience	Alternative, national
Federated learning workshops at ML conferences	FL robustness and privacy audience	If extended with stronger attacks

Call dates and page limits change every year; the team will confirm the current call for papers before choosing the final venue.

Paper outline

- 1. Introduction: cross-border AML, legal constraints, problem statement.
- 2. Background and related work: FL, DP, robust aggregation, blockchain-based FL, FL for financial crime.
- 3. Threat model and requirements.
- 4. SENTINEL-FL design: protocol, privacy, registry contract, ledger and anchoring.
- 5. Implementation.
- 6. Evaluation: the five scenarios, overhead.
- 7. Discussion: secure aggregation versus accountability, limits.
- 8. Conclusion.

Work needed before submission

- Re-run all experiments on IBM AMLworld (HI-Small and LI-Small) and report results on the Flower + PyTorch + Opacus engine.
- Add adaptive and colluding poisoning attacks, and a stronger privacy attack (shadow-model membership inference).
- Short related-work comparison against existing federated AML approaches; anonymise the repository for double-blind review if required.